

In Table S1-S5, all costs are reported in constant 2025 USD, with a project lifespan of 20 years and a discount rate of 3-8% applied to reflect financial uncertainty. Water volumes are consistently expressed in Mm³/year, and energy values in kWh or GWh. Cost ranges represent variability across geographic, technological, and market conditions based on literature and industry benchmarks. Key system parameters, including recharge rates, storage capacity, and energy consumption, are derived from the model developed in this study. Economic benefits, such as avoided costs, forest and ecosystem services, and carbon storage, are estimated using simplified relationships and should be interpreted as indicative ranges. Energy costs assume a hybrid solar-gas supply, while filtration media volumes and replacement cycles are based on assumed porosity and operational lifetimes of 2-4 years. Data sources include peer-reviewed studies such as (Brander et al., 2023; Dennis et al., 2024; Kiddu & Nkwasiwe, 2025; Leong, 2025; Martin et al., 2014; Sitek et al., 2025; Stegarescu & do Rosario Partidario, 2014) etc., Ampac USA (Ampac Water Services, (Gustafson, 2026)), Sustainable Atlas, Clean Water Store, international technical reports from organizations including the Food and Agriculture Organization (FAO), World Bank Group, and International Renewable Energy Agency, as well as other engineering and energy system assessments. Additional general reference values, such as typical ranges for material properties or unit costs, were cross-checked using open-access repositories such as Wikipedia where appropriate. All values were harmonized to ensure internal consistency in units, temporal resolution, and economic assumptions for subsequent analysis, and are subject to inherent uncertainty.

Table S1: Model Parameters

S. No.	Parameter	Symbol	Unit	Value	Remark
1	Project lifespan	N	years	20	Scenario
2	Discount rate	r	%	3-8	Sensitivity range
3	Time step	t	year	1-20	Annual
4	Water production rate	Q	Mm ³ /year	8.54	Computed
5	Total dissolved solids	TDS	Mg/L	334	Feedwater quality
6	Pump efficiency	η	%	70	Assumed
7	Water demand factor	α		0.8	Estimated
8	Energy cost	C _{energy}	\$/kWh	0.043-0.10	Estimated for solar-diesel

Table S2: Capital Expenditure (CAPEX)

S. No.	Component	Unit	Value (USD)	Remark
1	Water well (10 wells, each 12m deep)	USD	30,000-150,000	Estimated at 3,000-15,000 per well
2	Surface facilities	USD	50,000-150,000	Fencing, pump/powerhouse, control systems, storage tanks, access roads etc.
3	Treatment system	USD	10,000-30,000	Inlet structures, water quality test etc.
4	Installation & engineering	USD	40,000-120,000	Flow meters, IoT sensors, valves, distribution etc.
5	Permits & fees	USD	50-3000	Government approved license
6	Recharge basin, farmland, green areas, forest reserve, eco zones	USD	1.7M-5.8M	Estimated for 3500 x 4200 m ²
7	Tree seedlings	USD	76,800-80,000	Estimated for 100,000 trees at 0.77/seedling
8	Labour, logistics, consultations	USD	100,000-100,000	Excluding operations
Total		USD	2.01M-6.43M	

Table S3: Operating Expenditure (OPEX)

S. No.	Component	Unit	Value (USD)	Remark
1	Energy consumption	kWh/m ³	0.03-0.40	Estimated cost for water pumping
2	Energy cost	USD/year	Computed	From $Q \times \text{energy} \times \text{price}$
3	Labour & maintenance	USD/year	2-5% CAPEX	38,000-311,000
4	Monitoring	USD/year	50-400	Passive system, cheap/automated IoT
5	Media replacement	USD/year	13,000-72,000	Biochar, zeolite and activated carbon
6	Media lifespan	years	2-4	Gross estimate for biochar, zeolite and AC
7	Biochar cost, BC	USD/m ³	0.10-0.50	Computed as $Q \times BC$
8	Zeolite cost, ZC	USD/m ³	0.10-0.50	Computed as $Q \times ZC$
9	Activated Carbon, AC)	USD/m ³	0.28-0.30	Computed as $Q \times AC$
Total		USD/year	64,250-735,900	

Table S4: System Performance and Mass Balance

S. No.	Parameter	Unit	Value (USD)	Remark
1	Total aquifer volume	Mm ³	29.4	Based on basin/land area and porosity of 20%
2	Total storage capacity	Mm ³ /year	18.78	From the infiltration rate of 0.01 m/day, covering 35% of total area of the basin
3	Recharge volume	Mm ³ /year	12.21	65% of total storage capacity
4	Water supply/withdrawal, Q	Mm ³ /year	8.54	Estimated from 70% yield of total storage
5	Domestic/industrial, D/I	Mm ³ /year	5.98	70% of total water supplied, Q
6	Irrigation/Afforestation, I/A	Mm ³ /year	2.56	30% of total water supplied, Q

Table S5: Economic Benefit Summary (Annual)

S. No.	Benefit	Equation	Value (USD) per year	Remark
1	Direct Water sales, DWS	$Q \times P$	3.89M-8.72M	Unit price of \$0.5-\$1.2/m ³ - D/I and \$0.35-\$0.6/m ³ I/A
2	Avoided desalination + brine disposal, AC_t	$Q \times (C_{SWRO} + C_{brine})$	9.40M-64.09M	Cost_SWRO = \$0.8-2.5/m ³ ; Cost_brine = \$0.3-5.0/m ³ , marginal supply cost replacement
3	Water quality improvement, WQ_t	$Q \times C_Q$	0.34M-0.85M	C_Q = \$0.04-0.10/m ³
4	Energy savings, ES_t	$Q \times \Delta E \times price$	0.77M-3.82M	$R_{SWRO} - E_{OPEX}$: 2.5-4.5 – 0.03-0.4 kWh/m ³ = 1.9-382 GWh/year, carbon/emission reduction, NOT cost saving
5	Groundwater benefit, ST_t	$Rch \times V_{storage}$	1.22M-6.11M	$V_{storage}$ = 0.10-0.50, resilient/security
6	Agriculture benefit, AG_t	$Q \times P_{irrig}$	0.51M-3.85M	P_{irrig} = \$0.2-1.5/m ³ , yield improvement
7	Carbon storage, CO_{2t}	$Q \times R_{CO_2} \times V_{Carbon}$	0.64M-0.4.49M	R_{CO_2} = 1.5-3.5 kg/m ³ , V_{CO_2} = \$50-150/ton, storage value
8	Forest/Ecosystem services, ECO_t	$Area \times V_{ECO}$	0.06M-12.05M	V_{ECO} = 0.01-2.0, 41% of total land area

Some Selected Reference

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